



STM32G431x6 STM32G431x8 STM32G431xB

Arm[®] Cortex[®]-M4 32-bit MCU+FPU, 170 MHz / 213 DMIPS,
up to 128 KB Flash, 32 KB SRAM, rich analog, math accelerator

Datasheet - production data

Features

- Core: Arm[®] 32-bit Cortex[®]-M4 CPU with FPU, Adaptive real-time accelerator (ART Accelerator) allowing 0-wait-state execution from Flash memory, frequency up to 170 MHz with 213 DMIPS, MPU, DSP instructions
- Operating conditions:
 - V_{DD} , V_{DDA} voltage range: 1.71 V to 3.6 V
- Mathematical hardware accelerators
 - CORDIC for trigonometric functions acceleration
 - FMAC: filter mathematical accelerator
- Memories
 - 128 Kbytes of Flash memory with ECC support, proprietary code readout protection (PCROP), securable memory area, 1 Kbyte OTP
 - 22 Kbytes of SRAM, with hardware parity check implemented on the first 16 Kbytes
 - Routine booster: 10 Kbytes of SRAM on instruction and data bus, with hardware parity check (CCM SRAM)
- Reset and supply management
 - Power-on/power-down reset (POR/PDR/BOR)
 - Programmable voltage detector (PVD)
 - Low-power modes: sleep, stop, standby and shutdown
 - V_{BAT} supply for RTC and backup registers
- Clock management
 - 4 to 48 MHz crystal oscillator
 - 32 kHz oscillator with calibration
 - Internal 16 MHz RC with PLL option ($\pm 1\%$)
 - Internal 32 kHz RC oscillator ($\pm 5\%$)
- Up to 86 fast I/Os
 - All mappable on external interrupt vectors
 - Several I/Os with 5 V tolerant capability
- Interconnect matrix
- 12-channel DMA controller
- 2 x ADCs 0.25 μ s (up to 23 channels). Resolution up to 16-bit with hardware oversampling, 0 to 3.6 V conversion range
- 4 x 12-bit DAC channels
 - 2 x buffered external channels 1 MSPS
 - 2 x unbuffered internal channels 15 MSPS
- 4 x ultra-fast rail-to-rail analog comparators
- 3 x operational amplifiers that can be used in PGA mode, all terminals accessible
- Internal voltage reference buffer (VREFBUF) supporting three output voltages (2.048 V, 2.5 V, 2.95 V)
- 14 timers:
 - 1 x 32-bit timer and 2 x 16-bit timers with up to four IC/OC/PWM or pulse counter and quadrature (incremental) encoder input
 - 2 x 16-bit 8-channel advanced motor control timers, with up to 8 x PWM channels, dead time generation and emergency stop
 - 1 x 16-bit timer with 2 x IC/OCs, one OCN/PWM, dead time generation and emergency stop
 - 2 x 16-bit timers with IC/OC/OCN/PWM, dead time generation and emergency stop
 - 2 x watchdog timers (independent, window)
 - 1 x SysTick timer: 24-bit downcounter
 - 2 x 16-bit basic timers
 - 1 x low-power timer

